

Multi-fidelity validation summary | 6 paired cases, 5 s each

Backend	TP MAPE	TTFT MAPE	Total process wall time	Speedup vs. CHIPSIM
CHIPSIM (reference)	0.00%	0.00%	69.00 h	1.00×
HYDRA-Packet	7.22%	0.67%	4.11 h	16.80×
HYDRA-Analytic	12.71%	0.52%	66.46 s	3,737.57×

CHIPSIM is the comparison reference. MAPE weights each design point equally.
Speedup uses summed process wall times; these sums are distinct from parallel elapsed time.